

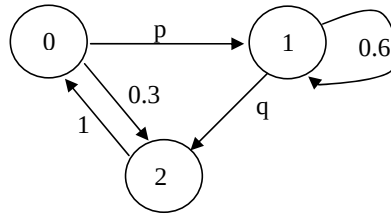


Semester II Examinations 2011-2012

Exam Codes	3BA1, 4BA4, 3BME1, 3BS4, 3BS9, 3BS10, 4BS4, 4CS2/3FM2
Exam	3 rd and 4 th Arts, 3 rd and 4 th Science/3 rd Science
Module	Modelling II/Mathematical Modelling
Module Code	MP307/MP391
External Examiner	Prof. Giuseppe Saccomandi
Internal Examiners	Prof. Michel Destrade Dr. Michael Tuite
Instructions	For full marks answer question 1 of Section A and TWO questions in Section B correctly. Each question is worth 20 Marks
Duration	TWO HOURS
No. of Pages	Four pages
Discipline	Applied Mathematics
Requirements	Release to Library: Yes Statistical Tables: Yes - Log Tables

Section A - Answer all questions

1. (a) **[5 Marks]** Customer arrivals in a queue are described by a Poisson process with parameter α . Show that the probability for k customers to arrive in time t is given by the Poisson distribution with parameter αt . Hence find the average number of arrivals per unit time.
- (b) **[5 Marks]** Consider the following Markov Process.



- (i) Find the transition matrix P and the probabilities p and q .
 - (ii) Is the Markov process ergodic?
 - (iii) Find the equilibrium probabilities.
- (c) **[5 Marks]** Consider the non-linear population model

$$P_{n+1} = R(P_n)P_n$$

where

$$R(P) = \frac{rP}{1 + \left(\frac{P}{K}\right)^2}, \quad r, K > 0.$$

Find the fixed points for this model and show for what initial populations P_0 the population will stabilize and when it will die out.

- (d) **[5 Marks]** Consider the linear control system

$$\begin{aligned} \frac{dx_1}{dt} &= x_2 + u, \\ \frac{dx_2}{dt} &= x_1 + x_2, \\ y &= x_1, \end{aligned}$$

where x_1 and x_2 represent the state, $u \in \mathbb{R}$ is the input and $y \in \mathbb{R}$ is the output.

- (i) Is this system observable?
- (ii) Is this system reachable?

Section B - Answer any **two** questions.

2. A queue of maximum size r has a Poisson arrival pattern with parameter α_k and Poisson servicing pattern with parameter β_k , where k is the queue size with $0 \leq k \leq r$.

- (a) **[8 Marks]** Explain why $\alpha_r = \beta_0 = 0$. Assuming the system is ergodic show that the equilibrium probabilities are for $0 < k \leq r$ given by

$$\pi_k = \rho_k \rho_{k-1} \dots \rho_1 \pi_0$$

where $\rho_k = \alpha_{k-1} / \beta_k$.

- (b) **[6 Marks]** Consider a telephone system exchange with r operators. Assuming the probability of a call arriving is independent of the queue size, find the equilibrium probabilities.
- (c) **[6 Marks]** An exchange consists of two operators, each of which can deal with 1 customer enquiry per 2 minutes on average. If 60 calls per hour arrive on average show that the equilibrium probability for all lines to be busy is 0.4. What would be the effect of introducing a third operator?
3. Consider an ecological system with a prey population of size $x(t)$ and a predator population of size $y(t)$, which satisfy the differential equations

$$\begin{aligned} \frac{dx}{dt} &= (1 - 2y)x, \\ \frac{dy}{dt} &= (-1 + 2x)y \end{aligned}$$

with initial populations $x(0) = x_0 > 0$ and $y(0) = y_0 > 0$.

- (a) **[4 Marks]** Give an interpretation for the various terms in these equations.
- (b) **[8 Marks]** Show that the equilibrium points are $(x, y) = (0, 0)$ and $(x, y) = (\frac{1}{2}, \frac{1}{2})$. For small deviations away from the equilibrium point $(x, y) = (\frac{1}{2}, \frac{1}{2})$, show that the system is periodic with period 2π .
- (c) **[8 Marks]** Show that in general the solution satisfies

$$K = \frac{e^{2(x+y)}}{xy}$$

for some constant K . Sketch a plot of x versus y and comment on the typical behaviour of this system in an application of this model.

4. A two-compartment model for drug delivery can be modelled as

$$\begin{pmatrix} \frac{dc_1}{dt} \\ \frac{dc_2}{dt} \end{pmatrix} = \begin{pmatrix} -p_0 - p_1 & p_1 \\ p_2 & -p_2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} b_0 \\ 0 \end{pmatrix} u, \quad y = c_2,$$

where c_1 and c_2 are the concentrations of the drug in each compartment, p_0, p_1, p_2 and b_0 are rate constants, u is the flow rate of the drug into compartment 1 and y is the concentration of the drug in compartment 2.

- (a) **[4 Marks]** Draw a schematic diagram of this system including concentrations, flow directions and rate parameters.
- (b) **[6 Marks]** Investigate for what values of p_0, p_1 and p_2 this system is observable.
- (c) **[10 Marks]** Design a control function

$$u = -Kc + k_r r, \quad \text{for } K = \begin{pmatrix} k_1 & k_2 \end{pmatrix}, \quad c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

where k_1, k_2, k_r are the control parameters, such that the desired output is $r = 2$ and the closed loop eigenvalues for the system satisfies $\lambda_1 = -2$ and $\lambda_2 = -1$.